SWINBURNE UNIVERSITY OF TECHNOLOGY

TNE20003 – Internet and Cybersecurity for Engineering Applications

Portfolio Task – Lab 5 Distinction Task

Aims:

- Explain & understand NAT at deeper level

Preparation:

- View "NAT & DHCP"

Due Date:

- Your Task will be assessed via an online quiz. You must score the required minimum to pass the test. You will be allowed a number of attempts to pass the test at the grade level you attempt. You are encouraged to complete the test at the end of the lab but if you do not, you must complete it before your next lab class.

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Task 1.

Build the network provided in figure 1 with Cisco Packet Tracer

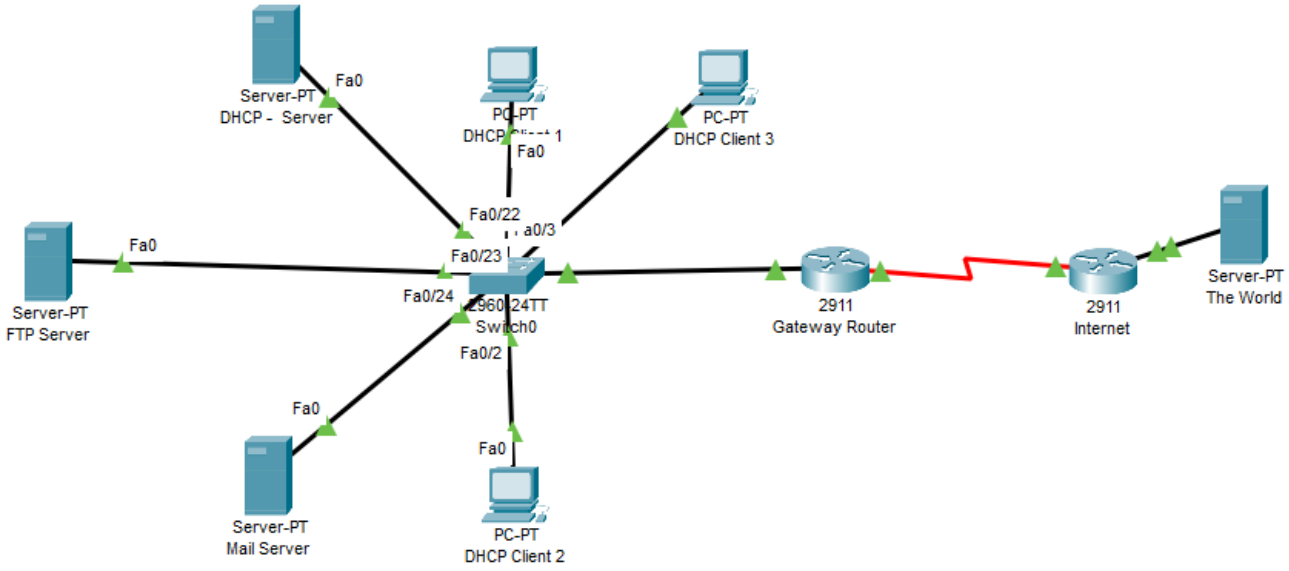


Figure 1

To attempt the D task you must have completed the P and C tasks. Use the network you built and tested in the Credit task for lab 5 to test and answer the following questions.

Task 2.

1. You have implemented NAT successfully in the Pass task.

    a. Now generate some traffic by pinging from DHCP Client 3 to **The World** server located at 136.186.100.2. What happens?

    b. Are there any address translations on the **Gateway** Router# prompt? Explain

    c. Ping **The World** server located at 136.186.100.2 from the **ftp** server. What happens?

d. How far does the ping from the **ftp** server propagate through the network?

e. Were there any translations on the Gateway router? Explain why/why not.

f. Change the IP address of DHCP Client 3 to a static address of 192.168.1.80. Ping **The World** server located at 136.186.100.2 from this **PC**. What happens?

g. Part of your output should be as shown in the picture below:

```
Router# sh ip nat tran
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.8.8.1:5       192.168.1.80:5     136.186.100.2:5    136.186.100.2:5
```

h. What does the **:5** represent?

i. Now ping from DHCP Client 2 and see the translations. What do you see?

j. What does inside local mean?

k. Change the IP address of DHCP Client 3 to a static address of 192.168.1.180. Ping **The World** server located at 136.186.100.2 from this **PC**. What happens?

l. Were there any translations on the **Gateway** router? Explain why/why not.

~~~~~ End of Lab ~~~~~